

SRS PART 2

Analysis Models AND System Features(5.1-5.6)

4. Analysis Models

The following analysis models describe the major interactions and processing flow of the proposed system. The models are intentionally kept simple so that the relationship between users, system components, and the verification process is clear.

4.1 Use Case Diagram

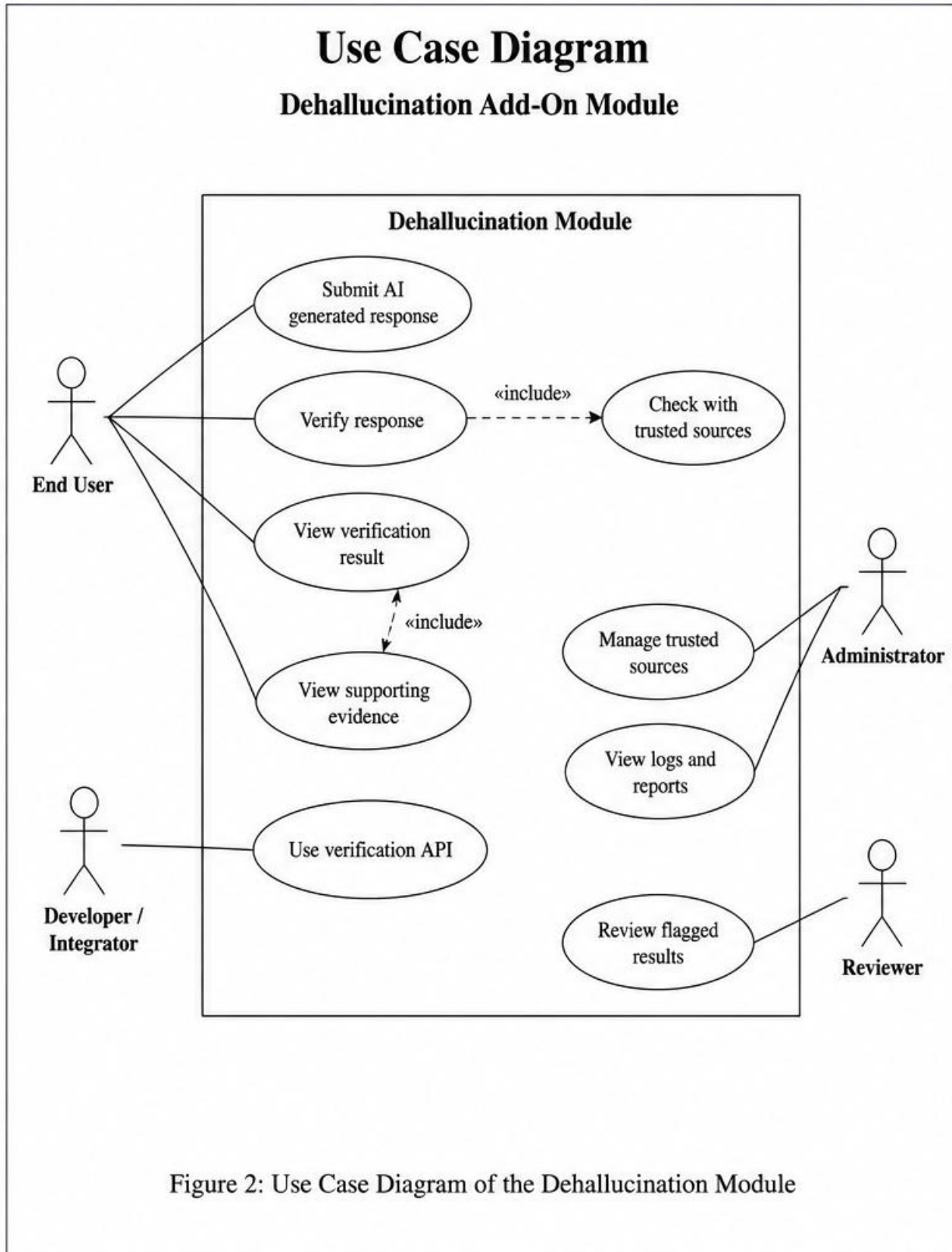
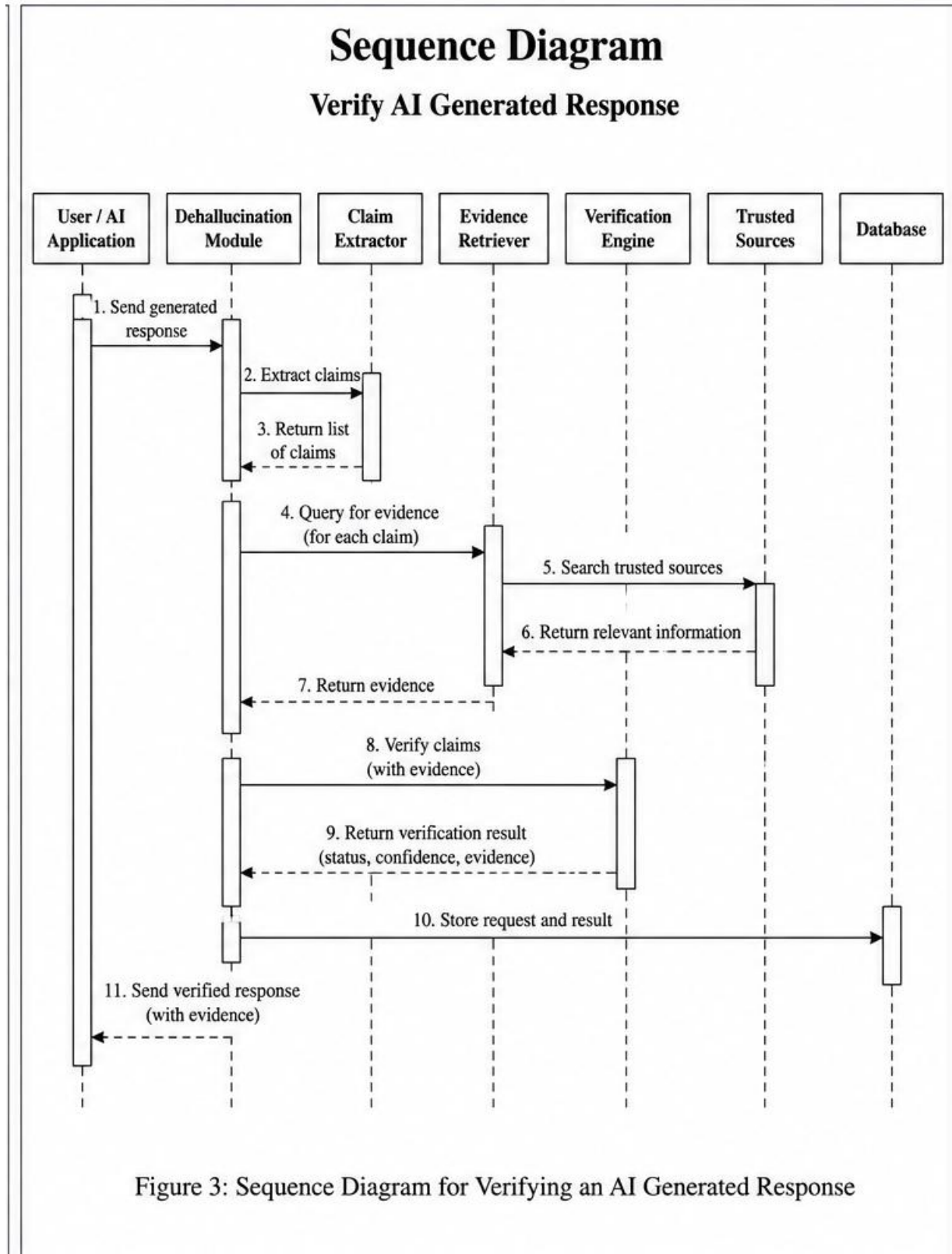


Figure 2: Use Case Diagram of the Dehallucination Module

The main actors are the End User, Developer/Integrator, Administrator, and Reviewer. The End User submits and verifies responses and views results/evidence. The Developer/Integrator uses the

verification API. The Administrator manages trusted sources and logs. The Reviewer examines flagged results.

4.2 Sequence Diagram



The sequence begins when the AI application sends generated text to the dehallucination module. Claims are extracted, evidence is requested from trusted sources, and the verification engine compares claims with evidence. The result is stored and returned as a verified response with evidence and verification information.

5. System Features

5.1 Response Input and Validation

Description and Priority: Receives an AI-generated response and checks whether the input is valid before starting verification. Priority: High.

Stimulus/Response Sequences:

1. User/application submits generated response.
2. System validates the request and preprocesses the text.
3. If valid, the system starts claim extraction.
4. If invalid, the system returns an error without performing verification.

Functional Requirements:

FR-01: The system shall accept an AI-generated text response for verification.

FR-02: The system shall validate that the submitted response is present and in the supported format.

FR-03: The system shall reject empty, malformed, or unsupported input and return an appropriate error message.

FR-04: The system shall preprocess valid text before claim extraction.

5.2 Claim Extraction

Description and Priority: Identifies factual statements or claims in the generated response that can be checked against evidence. Priority: High.

Stimulus/Response Sequences:

5. Validated response is passed to the claim extraction component.
6. The component identifies candidate factual claims.
7. The extracted claims are returned for verification.
8. If no verifiable claim is found, the system shall report that result.

Functional Requirements:

FR-05: The system shall identify factual claims from a valid generated response.

FR-06: The system shall maintain the association between each extracted claim and its original response.

FR-07: The system shall handle responses for which no verifiable factual claim is identified.

5.3 Trusted Evidence Retrieval

Description and Priority: Searches configured trusted sources and retrieves information relevant to each claim. Priority: High.

Stimulus/Response Sequences:

9. System creates a search query for a claim.
10. The retrieval component sends the query to trusted sources.
11. Relevant information is returned.
12. Unavailable sources or failed retrievals are recorded.

Functional Requirements:

FR-08: The system shall generate or accept a search query for each claim requiring evidence.

FR-09: The system shall retrieve relevant evidence from configured trusted sources.

FR-10: The system shall record source information associated with retrieved evidence.

FR-11: The system shall report retrieval failure when evidence cannot be obtained.

5.4 Fact Verification

Description and Priority: Compares extracted claims with retrieved evidence and assigns a verification status. Priority: High.

Stimulus/Response Sequences:

13. Claims and evidence are provided to the verification engine.
14. The engine compares the claim with the evidence.
15. Each claim is assigned a status such as supported, contradicted, or uncertain.
16. The result is passed to result generation.

Functional Requirements:

FR-12: The system shall compare each verifiable claim with retrieved evidence.

FR-13: The system shall classify a claim as supported, contradicted, or uncertain according to the implemented verification method.

FR-14: The system shall preserve the evidence used for each verification decision.

FR-15: The system shall not mark a claim as verified when the verification result is uncertain.

5.5 Confidence Scoring

Description and Priority: Provides confidence information for verification results so that users can distinguish strong results from uncertain ones. Priority: Medium.

Stimulus/Response Sequences:

17. Verification result is produced.
18. The scoring component calculates or assigns a confidence value.
19. The confidence information is attached to the claim/result.
20. Low-confidence results may be flagged for review.

Functional Requirements:

FR-16: The system shall assign a confidence score or confidence level to a verification result.

FR-17: The system shall associate confidence information with the corresponding claim.

FR-18: The system shall flag results that fall below the configured confidence threshold, if a threshold is used.

5.6 Hallucination Detection and Response Processing

Description and Priority: Uses claim verification results to identify potentially hallucinated content and decide how the final response should be presented. Priority: High.

Stimulus/Response Sequences:

21. System receives claim-level verification results.
22. Unsupported or contradicted claims are identified.
23. The system decides whether to flag, remove, qualify, or modify affected content according to the configured project rule.
24. A final response is generated.

Functional Requirements:

FR-19: The system shall identify response content associated with contradicted or unsupported claims.

FR-20: The system shall mark potentially hallucinated content for user awareness.

FR-21: The system shall generate a verified or modified response based on the verification results.

FR-22: The system shall retain the original generated response for comparison when logging is enabled.